The Intensive Margin of Small-Firm Protection: Bidding, Rents, and the Fiscal Cost of Public Procurement Set-Asides*

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Abstract

Using within-auction bid-distribution moments around a discrete policy cutoff, I show that the price effect of SME set-asides is not a pure headcount effect and is most consistent with a substantial within-auction bidding channel. When Brazil's March 2018 reinterpretation extended Sao Paulo's SME-only regime to medical supplies—the only previously exempt product group—the winning bid moved roughly six percentage points closer to the buyer's reference price, an effect that persists when firm count is held fixed at two, three, or five-plus. A Gelbach decomposition assigns only six percent of the 10–11% price increase (7–10% real) to entry and winner composition; ninety-four percent is unexplained by these observable channels, and direct phase-2 bid-moment evidence is consistent with within-auction bidder behavior as an important source. Identification uses a two-way fixed-effects staggered DiD (item and month FE) with 76 never-treated controls, robust to the Callaway-Sant'Anna, Sun-Abraham, and Goodman-Bacon estimators and to a formal spillover test. The static fiscal cost for Group 65 alone is R$50–74 million (US$14–21 million) over 18 months; 92% of procurement value concentrates in the top quartile, so a mechanical value-threshold exemption recovers 90% of the accounting cost while preserving the SME preference on 75% of items. The distortion is roughly twice as large

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An additional bidder is worth more to the auctioneer than any sophisticated bidding mechanism. — Bulow and Klemperer (1996), paraphrased.

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(coefficient −0.156 vs. −0.075) in CMED price-regulated pharmaceuticals, exposing a perverse interaction between two Brazilian regulatory regimes.

Keywords: public procurement, SMEs, policy costs, restricted public tenders.

JEL No.: H32, H57, L26, L53, O12, R12.

1. Introduction

When a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? Four decades of research on bid preferences and set-asides—from Marion (2007) through Krasnokutskaya and Seim (2011), Athey et al. (2013), and Nakabayashi (2013), covering billions of dollars in policy evaluation—have not answered the question directly. The reason is data: existing strategies back out the bidding channel from cross-sectional variation in winning prices alone, a design that conflates entry with aggressiveness by construction. The bid distribution itself has remained a black box.

The answer matters. Set-asides for small and medium-sized enterprises (SMEs) are the most widespread procurement-preference instrument worldwide: the European Union channels a large share of public contract value to SMEs under similar arrangements (PwC, 2014; Loader, 2013, 2015), OECD and SIGMA surveys document comparable regimes across more than one hundred jurisdictions (OECD, 2018; SIGMA, 2016; Hoekman and Tas, 2020), and Brazilian federal law mandates exclusive SME tenders for items valued below R\$80,000 (US\$22,900),[1] covering approximately 97% of firms representing half of formal employment (Bastos et al., 2018; Thai, 2017; Bosio et al., 2022). A substantial literature shows that these policies boost firm growth, employment, and

[1] All USD figures in this paper use a reference exchange rate of R\$3.50 per US dollar, the average of the BCB monthly closing rate over the September 2016–August 2019 sample period. The 2016–2019 annual averages were R\$3.49 (2016), R\$3.19 (2017), R\$3.65 (2018) and R\$3.88 (2019).

supplier-network expansion (Mel et al., 2008; Freedman, 2013; Szerman, 2023; Cardoza et al., 2016; Fafchamps et al., 2014; McKenzie and Woodruff, 2015; Ferraz et al., 2016; Cabral, 2017). But the design of the fix depends on the channel. If the price premium runs through the extensive margin, expanding the SME roster—through subsidies, registration relief, or capacity-building—can lower the fiscal cost without weakening the preference. If instead it runs through the intensive margin, no roster expansion will help: the only lever is to shrink the *scope* of the rule itself. Policy design hinges on the answer—for the procurement officers in Brazil's 5,570 municipalities implementing Law 14,133/2021, for the 130 countries with SME-preference regimes in place, and for development economists wrestling with the trade-off between competitive efficiency and distributional objectives in public spending.

Theory points both ways. Bulow and Klemperer (1996) show that, across a wide class of auction environments, an additional bidder is worth more to the auctioneer than any sophisticated mechanism—a headline result for the extensive margin. Krasnokutskaya and Seim (2011) formalize bid preferences in the procurement context and show that both margins are operative. Which dominates quantitatively is an empirical question, and outside the US the evidence for SME set-asides is essentially absent (Nakabayashi, 2013; Szucs, 2024; Decarolis et al., 2025).

This paper shows that the price effect is not a pure headcount effect and that an important within-auction bidding channel—the *intensive margin*—is the most plausible reading of what else is going on. Using phase-2 electronic-auction bid-distribution moments from 3.7 million observations on Sao Paulo's BEC platform, I exploit a March 2018 reinterpretation that extended the state's SME-only regime to medical supplies (Group 65), the only previously exempt product group. The switch was driven by a legalistic isonomy opinion from a separate oversight agency (PGE-SP), plausibly exogenous to Group 65 procurement outcomes. Group 65 is the lone treated cohort and the 76 other groups—subject to SME-only since the 2014 federal mandate—are never-treated controls within the sample window, so identification comes from a clean treated-vs-never-treated comparison, free of the "forbidden" implicit comparisons that motivate the modern staggered-DiD literature (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024); the Kim and Lee (2019) DiDiR framework delivers the same estimand.

Imposing the SME-only rule on Group 65 raises procurement prices by 10–11% over 18 months (13–14% at 6 months, 7–10% in IPCA-deflated real terms), reduces participating firms by 10–11%, and cuts valid bids by 7–8%. Winning firms under open tenders sit 6–14 km further from buyers. The mechanism shows up directly in phase-2 bid moments: the winning bid reaches about six percentage points deeper below the reference under open tenders, an effect that survives conditioning on exactly two, three, or five-plus firms. The Gelbach (2016) decomposition corroborates: six percent of the price reduction runs through entry and winner composition, and 94% is unexplained by those channels, consistent with a within-auction bidding-behavior story. Pairing a discrete policy cutoff with within-auction bid-distribution data sidesteps the identification problems that have constrained cross-sectional studies of bid preferences (Marion, 2007; Krasnokutskaya and Seim, 2011; Athey et al., 2013).

Identification varies by outcome. The log-price effect is the core causal estimate: the price placebo is one-quarter to one-third of the headline (two of three cutoffs marginally significant). The log firms and log bids coefficients have pre-trend contamination that placebos and Sun-Abraham diagnostics do not resolve—they enter as suggestive supporting evidence, not as independently identified. The distance outcome is causally clean (null placebos); the bid-distribution moments driving the mechanism section are within-item and independent of the firms-bids pre-trend issue. Three staggered-DiD estimators (CS2021, Sun-Abraham, Goodman-Bacon) deliver the same sign on all four outcomes.

A formal cross-group spillover test bounds redirect bias at less than 1% of the headline, and RAIS-based firm-size, age, and CNAE controls leave the coefficient invariant. With a single switching cohort, classical permutation inference is underpowered ($p = 0.342$); the evidence for causality therefore rests on cross-estimator consistency rather than on that test.

The static fiscal cost of the SME-only rule for Group 65 alone is R$50–74 million (US$14–21 million) over the 18-month window (R$50 million real, R$74 million nominal), or 7–11% of the group's procurement value. The number is mechanical—a reduced-form price coefficient times pre-period procurement value—and ignores the completion, entry, and reference-price responses that a structural counterfactual would price. Two findings sharpen the policy implications. First, 92% of that procurement value is concentrated in the top value quartile, so a mechanical value-threshold exemption recovers 90% of the accounting cost while preserving the SME preference on 75% of items by count. SMEs lose access to the top-quartile contracts under this reform—so it is not strictly Pareto-improving, and a full welfare analysis would require modeling the behavioral responses—but even in static terms the reform substantially relaxes the efficiency-equity trade-off that motivates SME-only rules. Second, CMED-regulated pharmaceuticals (class 6531 within Group 65) bear more than twice the distortion of the non-pharma medical supplies: the CMED list-price cap does not shield procurement from the ME/EPP preference, consistent with contract prices lying below the cap and the restriction compressing competition within the residual slack. Two Brazilian regulatory regimes—SME procurement preferences and drug price controls—therefore interact perversely, raising public-health spending above what either instrument alone would deliver.

The paper makes four contributions. *First*, direct bid-distribution evidence on the intensive margin: using phase-2 bid moments, I show the winning bid moves about six percentage points further below the buyer's reference price under open tenders (conditional on firm count) and the runner-up-to-winner gap narrows by 2.7 percentage points, with the effect concentrated in auctions with five or more firms. Prior work has inferred the mechanism from winning-price variation alone (Marion, 2007; Athey et al., 2013; Krasnokutskaya and Seim, 2011; Nakabayashi, 2013); 94% of the price effect here lies outside measured entry and composition channels. *Second*, a sharp counterfactual policy-design result: because 92% of Group-65 procurement value concentrates in the top value quartile, a mechanical value-threshold exemption—applying the SME-only rule only below about R$7,600 (US$2,180)—recovers 90% of the accounting fiscal cost while preserving the preference on 75% of items by count. *Third*, a perverse interaction between two Brazilian price-regulation regimes: CMED-regulated pharmaceuticals (class 6531) bear more than twice the distortion of non-pharma medical supplies, because contract prices sit below the CMED cap and the SME restriction compresses competition within the residual slack—new to the procurement-preference literature (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2025; Colonnelli and Prem, 2022). *Fourth*, quasi-experimental estimates of SME set-aside costs from a large developing economy (R$50–74 million, US$14–21 million, over 18 months for a single product group), rare outside the US and large-firm Europe (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Bandiera et al., 2009; Best et al., 2023), using a single-cohort *staggered-DiD reversal* (Kim and Lee, 2019; Athey and Imbens, 2022) against 76 never-treated controls that is robust to Callaway-Sant'Anna, Sun-Abraham, Goodman-Bacon, a cross-group spillover test, and winner-level RAIS controls for firm size, age, and sector.

Section 2 provides the institutional background. Section 3 describes the data. Section 4 presents the empirical analysis, heterogeneity, robustness, and fiscal cost quantification. Section 5 concludes with policy implications.

2. Institutional Background

Brazil's federal SME statute (Complementary Law 123/2006, amended 2014) defines micro- and small enterprises by annual gross revenue—R$360,000 (US$103,000) and R$4,800,000 (US$1.37 million) ceilings, respectively—and mandates exclusive SME tenders for items valued at R$80,000 (US$22,900) or less. Public buyer units (PBUs)—federal, state, and municipal administrative entities subject to procurement rules—can avoid the SME-only default only by justifying the departure through a formal report to the audit courts, a costly and scrutinized process. The 2014 amendment converted what had been an option ("may") into a binding default ("must") and imposed these *opt-out costs*; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the subsequent years.

Since 2005, all public procurement in the state of Sao Paulo passes through Bolsa Eletrônica de Compras (BEC), a centralized electronic reverse-auction platform in which registered suppliers submit sealed bids (phase 1) followed by an iterative electronic auction (phases 2 onward) in which each participant may progressively lower its offer until the auction closes. BEC handled approximately R$13 billion (US$3.7 billion) of procurement in 2019 alone and has cumulatively moved more than R$105 billion (US$30 billion) since implementation, spanning 860,000 purchase offers and 4.8 million item-level transactions. BEC organizes its catalog into three tiers—group, class, and item—so that "medical, dental, and hospital equipment and supplies" form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).

Group 65 was a singular exception to the 2014 rule. Between August 2014 and February 2018, the state government and the Tribunal de Contas do Estado (TCE-SP) operated under a joint interpretation that medical supplies were strategic goods requiring open competition; during that window no Group-65 bid was restricted to SMEs. In November

2017, the Procuradoria-Geral do Estado (PGE-SP, a separate control agency) issued a legal opinion reversing the exception on isonomy grounds—the constitutional requirement that comparable cases be treated alike. The opinion argued that no substantive feature of medical supplies justified procedural distinctions from other product groups. Effective March 2018, opt-out costs extended to Group 65; SME-only adherence within the group jumped to 43% on average, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and often fail the three-eligible-SMEs requirement.

What matters for identification is that the policy change was triggered by a legalistic reinterpretation of the isonomy principle at PGE-SP, not by anything happening in the Group-65 procurement market. Public documentation turns up no sign that the decision was a response to price levels, competition patterns, or supplier complaints specific to Group 65 in 2017; the timing was set by the bureaucratic process inside PGE-SP and by when that office chose to communicate the opinion to PBUs. The change is therefore plausibly exogenous to the outcomes studied. This is the institutional basis for the identifying assumption that, absent the March-2018 reinterpretation, Group 65 would have tracked the contemporaneous trajectory of the 76 never-treated product groups.

3. Data

I use bidding-level administrative data on public procurement of common goods and services in the state of Sao Paulo, Brazil, from January 2016 to December 2019. All transactions flow through BEC, the state's centralized electronic procurement platform; SEFAZ/SP (the State Finance Secretariat) manages it and centralizes the bidding records. Because BEC covers the universe of standardized goods procurement in the state, the data are free from the selection concerns that arise with partial or voluntary coverage.

BEC serves 1,344 PBUs as regular buyers: state bureaus from the executive, legislative, and judicial branches of Sao Paulo, along with affiliated entities such as municipal governments and accredited private organizations. Between 2016 and 2019 these PBUs purchased 82,569 distinct items (goods) across 832,984 successful transactions.

BEC catalogs goods and services at three nested levels: group, class, and item. Health items sit in Group 65 (medical, dental, and hospital equipment and supplies). Within Group 65, class 6531 holds medications with or without ANVISA notification; item 110639, for example, is the drug “Furosemide 40 milligrams, coated tablets, units.” To make the policy shift concrete: in February 2018, under the open-tender regime, a Sao Paulo PBU bought 50,000 units of this same furosemide item in an electronic auction where ten firms pushed the bids down to 14 percentage points below the reference price, and the winning unit price cleared about 12% below reference. In October 2018, after the SME-only rule bound Group 65, a comparable furosemide purchase by the same PBU attracted three firms, phase-2 bids clustered within 3 percentage points of the reference, and the winning unit price came in just above reference.[2] The paper quantifies this pattern across the full Group-65 universe.

The data are organized by *purchase offer* (PO)—the electronic document a PBU issues to identify and quantify the goods or services it intends to buy. Each PO has a 22-character code and may list one or more items, but each item runs through its own purchase process. A unique item-level observation is therefore identified by the combination of PO and item codes (POI).

The central regressor, constructed from the item group code, is a binary indicator equal to one if the item belongs to Group 65 and zero otherwise. Group 65 is the *switched group*—the set of items affected by the March-2018 policy change—and accounts for nearly 27% of all purchases between 2016 and 2019. The remaining product groups form the control set.

[2] This stylized pair is illustrative rather than a case study; the average effects across the 101,386 class-6531 observations in the 18-month window are reported in Section 4.3 and the pharma decomposition table.

There are 76 product groups besides Group 65—the never-treated comparison set for the empirical design. Between 2016 and 2019, the most significant groups were groups 89 (food products), 75 (office supplies), 86 (computer products), and 79 (cleaning materials), representing 37.5% of the total purchases in this period. For each POI, there is information about item quantities, bid prices (winners and losers), the number of participant firms, the number of bids, whether the public tender was successful or not, the identification of the PBU, and firms' and PBUs' location, among other variables.

The empirical analysis tracks four outcomes. The *negotiated price* (in logs) is the final transacted price, available only for completed items. The *number of participant firms* (in logs) counts the distinct firms that submitted at least one bid, whether the item was ultimately purchased or not. The *number of valid bids* (in logs) counts the qualifying bids received per item. *Distance* is the kilometer distance from the winning firm's location to the buying PBU. Every regression controls for log quantity ordered and the type of tender (sealed bid vs. reverse auction), and includes item-level fixed effects that absorb all time-invariant heterogeneity across items—differences in markets, pricing norms, and typical supplier pools.

To rule out compositional bias, I restrict each regression to items that appear in both the pre- and post-periods within the relevant time window. Three nested windows (6, 12, and 18 months around the March-2018 cutoff) check that the estimates are stable as the sample expands. Price and distance regressions use the subsample of completed items (those with a winning bid); participation and bid regressions use all items, including those with no winner. Conditioning on completion raises a sample-selection concern—the treatment itself may affect whether an item is transacted—which the Lee bounds analysis in the Appendix addresses directly. Table 1 presents descriptive statistics for the key variables used in the empirical analysis, disaggregated by group (Group 65 vs. Others) and period (Pre vs. Post the policy shift in March 2018).

Table 1: Descriptive Statistics (18-month window)

	Group 65, Pre		Group 65, Post		Others, Pre		Others, Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Price (levels)	1,402.09	20,563.54	1,507.92	25,872.31	1,465.85	50,385.93	1,291.37	68,042.55
Log price	1.72	2.84	1.91	2.81	2.85	2.12	2.78	2.07
Number of firms	3.17	2.52	3.02	2.40	4.66	3.27	4.69	3.28
Log number of firms	0.88	0.73	0.84	0.71	1.30	0.72	1.30	0.72
Number of valid bids	11.10	15.49	10.99	16.51	10.13	15.74	10.47	16.87
Log number of valid bids	1.64	1.24	1.61	1.23	1.69	1.06	1.70	1.08
Distance (km)	238.54	270.42	215.08	239.63	164.78	182.23	170.28	186.07
N (completed / all)	65,606	77,206	66,822	85,181	246,277	287,500	271,011	323,836

Notes: Price and distance statistics computed on completed items (status = 1). Firm and bid statistics computed on all items.

4. Empirical Strategy and Results

4.1. Main Specification and Identification Strategy

The March 2018 policy change constitutes a single-cohort staggered difference-in-differences design. Group 65 enters the SME-only regime in March 2018, joining the 76 other product groups that had been subject to the restriction since August 2014. Within the 18-month estimation window, Group 65 is the sole treated cohort, and the 76 other groups are never-treated controls—they transitioned to SME-only before the sample window. Identification comes from comparing a single switching cohort to a stable never-treated comparison set, so the "forbidden" implicit comparisons that motivate the modern staggered-DiD literature (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Goodman-Bacon, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfoeuille, 2020; Roth et al., 2023) do not arise.[3] Table 2 restates the timeline in treatment-assignment form.

[3]Because the switch is *from* open tenders *to* the SME-only regime, the design is also a difference-in-differences in reverse (DiDiR) in the sense of Kim and Lee (2019); both framings yield the same estimand.

Table 2: Policy status by group and period

Groups	Before March 2018	After March 2018
Group 65 (treated cohort)	Open tenders	SME-only (opt-out costs > 0)
Other 76 groups (never-treated)	SME-only (opt-out costs > 0)	SME-only (opt-out costs > 0)

The estimating equation is

$$y_{pigt} = \eta_i + \lambda_t + \beta\,(g65_{pgt} \times \mathrm{Pre}_t) + x_{pigt}\,\delta + \varepsilon_{pigt} \tag{1}$$

where η_i is item fixed effects, λ_t is month fixed effects, $\mathrm{Pre}_t = 1$ before March 2018, $g65_{pgt} = 1$ for Group 65, and x_{pigt} collects sealed-bid and log-quantity controls. Standard errors are clustered at the item level. Sign convention: $\beta < 0$ for prices means Group 65 was cheaper under open tenders, so imposing SME-only raises prices by $|\beta|$ log points. For a single treated cohort this specification is algebraically equivalent to the Sun and Abraham (2021) interaction-weighted estimator, confirmed numerically in Section 4.4 alongside the Callaway and Sant'Anna (2021) group-time ATT.

The central identifying assumption is that, absent the March 2018 policy change, the trajectory of outcomes for Group 65 would have tracked the contemporaneous trajectory of the 76 control groups. Identification relies on parallel *changes* over time, which the item fixed effects isolate by comparing each item with itself across the regime cutoff. Anticipation is a second potential threat: the PGE-SP opinion dates from November 2017, four months before implementation. Front-loading under the soon-to-expire open-tender rule would add open-tender volume immediately before the cutoff, *attenuating* the estimated price gap—biasing toward zero. The DiD coefficients are in any case stable across the 6-, 12-, and 18-month windows: if anticipation mattered quantitatively, the 6-month estimate (which weights the months closest to the cutoff most heavily) would depart systematically from the 18-month estimate. It does not.

The parallel-trends assumption is partially assessed by an event study, $y_{pgt} = \eta_g +$

Semester$_t$ + $\sum_{k\neq-1}\beta_k\,(g65_{pgt}\times\mathbf{1}[t=k])+\mu_{pgt}$, in the spirit of Naritomi (2019) and Gallego et al. (2020). Figure 1 plots the log-price coefficients; the other outcomes are in the Appendix.

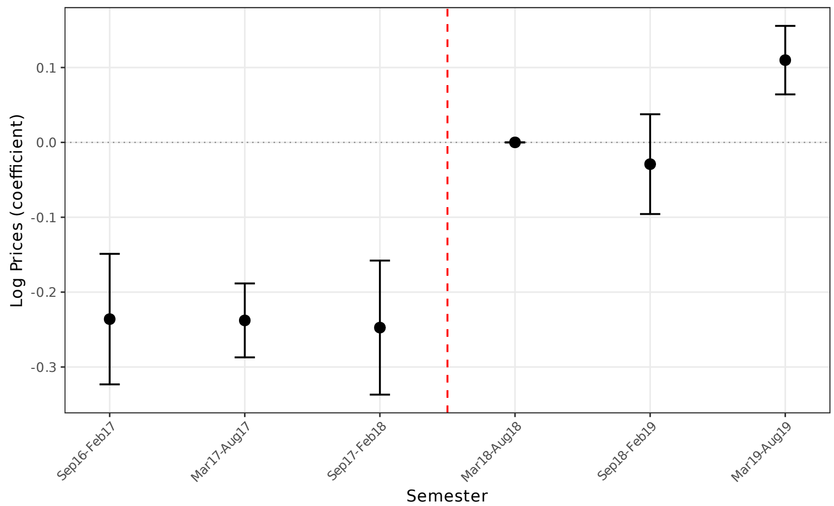

Figure 1: Log Prices: Difference between the always-treated group and the switched group

Even with only three post-policy semesters the pattern is clear: the gap narrows sharply once the rule binds; the first two post-semesters post near-zero coefficients, consistent with parallel trends. The final semester (March–August 2019) shows a positive, marginally significant coefficient—a partial reversal most consistent with PBU learning, given that the DiD estimates are stable across the 6-, 12-, and 18-month windows and the headline drops from −0.109 to −0.113 when the final semester is excluded (Online Appendix). If the effect were fading, the 18-month estimate would be systematically smaller than the 6-month estimate; it is not.

First stage: the policy shock binds. Before measuring outcome effects, it is worth verifying that the March-2018 reinterpretation actually changed who wins Group-65 items. Table 3 reports the DiDiR coefficient with the SME-winner indicator as dependent variable. Pre-treatment, 0.1% of Group-65 winners were SMEs—effectively none, because Group 65 operated under open tenders and larger distributors dominated. Post-treatment, the SME-winner share in Group 65 jumps to 39.1%. Control groups, which already operated under SME-only, move from 40.4% to 67.0% over the same window (a secular rise driven by factors unrelated to the cutoff). The within-item DiD estimate of the composition shift in Group 65 is +11 to +13 percentage points depending on window and fixed-effect specification, each significant at $p < 10^{-20}$. The legal reinterpretation measurably changes who wins; the estimated price, firm, and bid effects that follow are effects of *this* composition shift, not of an unbinding rule. Figure 2 visualizes the jump in event-study form.

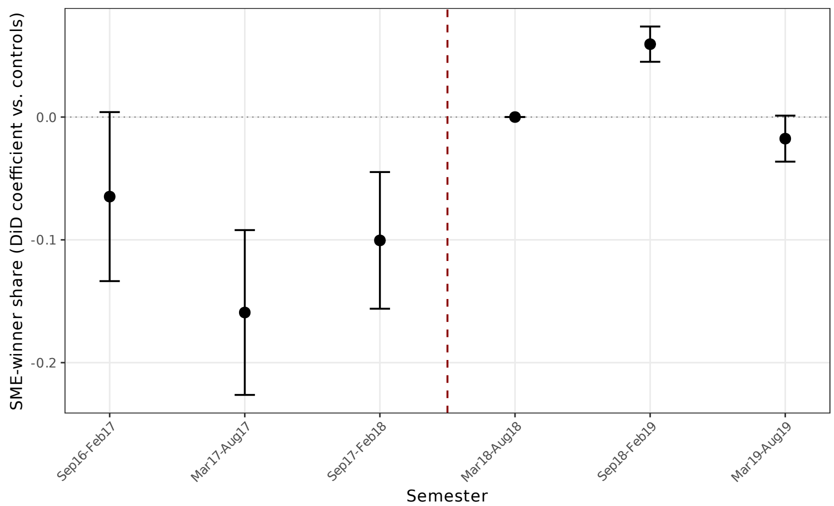

Figure 2: First stage: SME-winner share in Group 65 vs. controls, by semester. Semester 4 (March–August 2018) is the reference; negative pre-semester coefficients reflect the low SME share that Group 65 had under open tenders; the positive jump at semester 5 is the immediate effect of the March-2018 reinterpretation.

4.2. Results

Equation (1) is estimated on three nested windows—6, 12, and 18 months around the March 2018 cutoff—and under two specifications: item plus month fixed effects, and item plus month plus buyer-unit (PBU) fixed effects. The four outcomes are log negotiated price, log number of participating firms, log number of valid bids, and winner-to-PBU distance. Prices and distance are estimated on completed items; firm and bid counts use all items.

Table 3: First Stage: March-2018 Reinterpretation and the SME-Composition Shift in Group-65 Winners

	(1) 6m, item	(2) 6m, item+PBU	(3) 12m, item	(4) 12m, item+PBU	(5) 18m, item	(6) 18m, item+PBU
$g65 \times Pre$	-0.0743***	-0.0857***	-0.1274***	-0.1321***	-0.1001***	-0.1162***
	(0.0072)	(0.0072)	(0.0067)	(0.0066)	(0.0064)	(0.0063)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
Item FE	YES	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES	NO	YES

Notes: Dependent variable is an indicator for whether the winning firm was registered as ME or EPP. Sample is restricted to completed items (*oc_item_status* = 1). $g65 \times Pre$ identifies the Group-65 pre-treatment cell: a *negative* coefficient means Group-65 items had a lower SME-winner share *before* the reinterpretation than after, relative to the within-item average. Equivalently, the March 2018 cutoff raised the SME-winner share in Group 65 by about 11–12 percentage points above what item and PBU fixed effects alone would predict. Raw pre/post means: Group-65 pre = 0.1%, post = 39.1%; controls pre = 40.4%, post = 67.0% (controls also rise over time, so the DiD is smaller than the within-G65 jump). This is the *first stage* of the identification strategy: the legal reinterpretation measurably changes who wins Group-65 items. Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Table 4: Number of Participant Firms (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.2110***	0.2199***	0.1284***	0.1384***	0.0978***	0.1073***
	(0.0092)	(0.0093)	(0.0074)	(0.0075)	(0.0070)	(0.0070)
sealed-bids	0.3685***	0.3667***	0.3432***	0.3427***	0.3481***	0.3541***
	(0.0082)	(0.0091)	(0.0076)	(0.0088)	(0.0073)	(0.0082)
lquantity	0.1927***	0.1747***	0.1949***	0.1781***	0.1930***	0.1772***
	(0.0024)	(0.0023)	(0.0021)	(0.0020)	(0.0020)	(0.0020)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2145	0.1666	0.2110	0.1655	0.2076	0.1640
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

Table 4 shows a consistent rise in firm participation in Group 65 under open tenders: 23–25% higher in the 6-month window (coefficient 0.211), attenuating to 10–11% at 18 months (0.098 and 0.107), consistent with PBUs gradually adapting. The pattern for valid bids is similar (Table B.1, Appendix). These competition effects should be interpreted with caution: the placebo tests (Section 4.4) show significant pre-trends for both firms and bids from the secular rollout of SME restrictions across other groups. The price placebo passes cleanly, so the competition results read as suggestive supporting evidence rather than independently identified causal effects.

Table 5: Prices (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	-0.1417***	-0.1482***	-0.1079***	-0.1077***	-0.1087***	-0.1133***
	(0.0146)	(0.0144)	(0.0133)	(0.0129)	(0.0120)	(0.0117)
Sealed bids	-0.1409***	-0.2176***	-0.1380***	-0.1991***	-0.1455***	-0.2018***
	(0.0086)	(0.0130)	(0.0077)	(0.0122)	(0.0077)	(0.0114)
lquantity	-0.2616***	-0.2986***	-0.2634***	-0.2974***	-0.2617***	-0.2949***
	(0.0082)	(0.0089)	(0.0075)	(0.0080)	(0.0073)	(0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1919	0.2132	0.1943	0.2120	0.1947	0.2118
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$. Constant absorbed by fixed effects.

The 18-month price coefficients are -0.109 and -0.113, implying prices 10–11% lower under open tenders ($e^{\hat{\beta}} - 1$). The 6-month window yields -0.142 and -0.148 (13–14%), the sharpest immediate impact before PBUs adapt. These magnitudes are an order larger than the 3.8% preference penalty Marion (2007) reports for California highway contracts or the comparable trade-offs Krasnokutskaya and Seim (2011) document: the Brazilian rule excludes non-SME bidders entirely rather than applying a bid discount, and the thicker candidate pool for Brazilian medical procurement amplifies the intensive-margin channel that preference rules dampen. Exclusion-based set-asides deliver a qualitatively different welfare loss than bid-preference policies.

Winning suppliers for Group 65 were located 6–14 km further from PBUs before the cutoff (6-month 6.6 km, 12-month 13.8 km, 18-month 14.2 km in baseline specifications; 5.7–6.2 km with PBU FE; Table B.2, Appendix): the SME restriction does deliver geographic proximity.

Lee (2009) bounds correct for selection from the completion margin: after trimming 8.82% of the outcome distribution, the bounds are $[-0.046, -0.134]$, both significant (Table B.6). The Rambachan and Roth (2023) HonestDiD sensitivity (Appendix Figure A.4) places the price effect as robust only to modest parallel-trends violations ($\bar{M} \leq 0.1$), an honest caveat rather than reinforcement. Quantile DiD (Canay, 2011), causal forest (Athey et al., 2019; Wager and Athey, 2018), and synthetic control (Abadie et al., 2010; Ben-Michael et al., 2021) are reported in the Online Appendix; the quantile pattern is strongly negative at the low end ($\tau = 0.10$: -0.623) and fades toward zero at the upper tail, consistent with the Bulow and Klemperer (1996) prediction that additional bidders generate large surplus gains in thick markets. Item quantity is the dominant heterogeneity moderator in the causal forest (variable importance = 0.486).

4.3. Mechanism: The Intensive Margin of Bidding

The 10–11% price effect can come from two channels. The SME-only rule may reduce the number of firms willing to bid (extensive margin), or it may change how each participating firm bids—the level, the dispersion, and the strategic response to who else is in the room (intensive margin). This section presents three complementary pieces of evidence that together point to the intensive margin as the dominant mechanism. Each piece has limitations in isolation; their agreement is the argument.

Mediation decomposition. A Gelbach (2016) decomposition splits the total price effect across observable channels. Open tenders attract more firms—including larger non-SME suppliers—and that extra competition does pull prices down. But the entry and winner-composition channels together explain only about 6% of the total effect; 94% is left over once firm count and winner-SME status are accounted for (Figure 3). This 94% does not identify the intensive margin structurally: the residual bundles every unmeasured channel—bidder identities, reserve-price practices, buyer habits, endogenous auction completion, and more. What it *does* establish is that entry and winner composition alone cannot carry the price effect; whatever drives 94% of it lives somewhere else.

Figure 3: Gelbach mediation decomposition of the price effect
Share of the 10–11% 18-month price increase attributable to the two observable channels. Entry and winner composition together account for six percent of the effect. The remaining ninety-four percent is residual in the Gelbach sense: a magnitude not explained by firm count or winner-SME status, not a structural decomposition of the true mechanism.

Conditioning on firm count. Table 6 pushes further. Columns (1)–(3) sequentially add log firms and winner SME status to the baseline price regression; the treatment coefficient actually grows slightly in magnitude (from −0.109 to −0.115). The growth has a natural reading: controlling for log firms absorbs an attenuating channel—open tenders raise firm count, and more firms also (weakly) lower prices through a measured competition effect, so netting out that channel slightly magnifies the estimated direct impact of the regime switch. This reaffirms the Gelbach result that measured mediators absorb little of the effect. Columns (4)–(6) hold the extensive margin *exactly* constant by restricting the sample to items with exactly 2, exactly 3, or five-plus participating firms. The price effect survives at −0.093, −0.108, and −0.107 respectively, all significant at the 1% level. The caveat is that firm count is itself an equilibrium outcome, so "same N in both regimes" does not guarantee the same bidder identities or cost distribution. The result rules out the strongest version of the pure-headcount story—that the price effect is nothing but an entry count—while leaving richer compositional explanations on the table.

Table 6: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N≥5
$g65 \times Pre$	-0.1087***	-0.1186***	-0.1146***	-0.0926***	-0.1076***	-0.1073***
	(0.0120)	(0.0123)	(0.0123)	(0.0166)	(0.0176)	(0.0206)
Observations	649,714	648,616	648,616	105,957	102,831	260,530
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin approximately constant. The persistence of the price effect across all subsamples rules out a pure headcount explanation for the cost reduction under open tenders; it does not, on its own, identify the intensive margin structurally. Firm count is itself an equilibrium outcome, so "same N in both regimes" does not guarantee the same bidder identities or cost distribution. *** p<0.01, ** p<0.05, * p<0.1.

Direct evidence from phase-2 bid moments. Table 7 inspects the phase-2 bid distribution directly. 81% of completed items in the 18-month window have valid phase-2 moments, and valid-phase-2 coverage is not treatment-driven (DiD on the validity indicator is +0.002, SE 0.003). Under the two-way FE specification, the discount from reference widens by 5.8 pp under open tenders in the full sample, and by 6.5, 2.5, and

2.6 pp in the two-, three-, and five-plus-firm subsamples (Figure 4). Holding firm count approximately fixed, the winning bid still reaches further below the reference under open tenders. The standard deviation of phase-2 bids does not shrink monotonically once time FE are absorbed, so the paper's intensive-margin interpretation rests on the discount shift and the Gelbach residual rather than on SD compression. The bid moments are direct measurements of within-auction bidder behavior, not mediation residuals, and document that the bid distribution shifts rightward when the restriction binds.

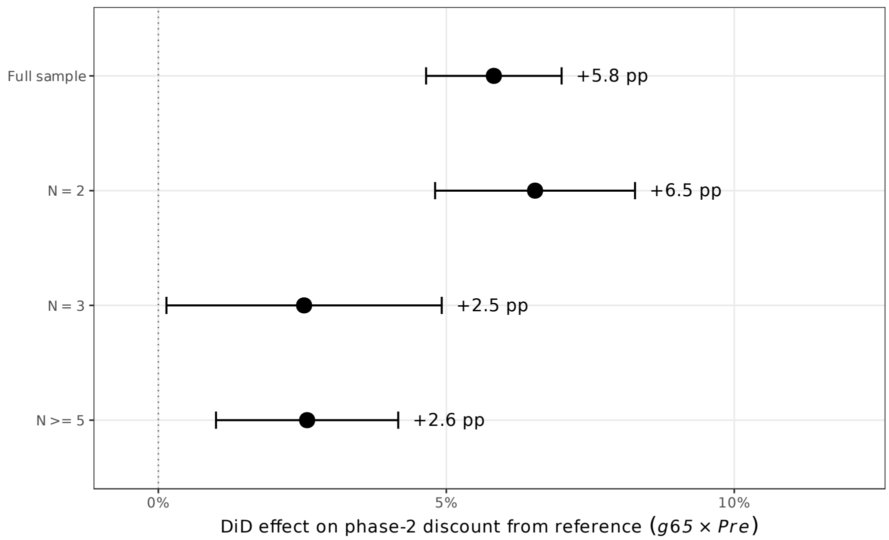

Figure 4: DiD effect on phase-2 winning-bid discount, by number of participating firms
Each point is the $g65 \times Pre$ coefficient from a separate DiD regression of the phase-2 winning-bid discount (fraction below reference price) on the treatment, item, month, and PBU fixed effects, and sealed-bid and log-quantity controls. Bars are 95% confidence intervals, clustered at the item level. Under open tenders the winning bid sits about six percentage points further below the reference price in the full sample, and the effect survives when the sample is restricted to items with exactly two, three, or five-plus participating firms (at smaller magnitudes)—i.e., when the extensive margin is held approximately constant.

Within-firm bid aggressiveness (firm fixed effects). Does the same firm bid differently, or does the SME-only rule merely shift the composition of winners? Table 8 adds winner-firm (CNPJ raiz) fixed effects to the discount regression. On the full sample, adding PBU and firm FE drops the coefficient from +0.072 to +0.046 (still highly significant): substantial within-firm response after removing bidder-composition variation.

Table 7: Bid-Level Aggressiveness: Phase-2 Moments (18-month window)

	Full sample	$N = 2$ firms	$N = 3$ firms	$N \geq 5$ firms
Discount from ref. (%)				
$g65 \times Pre$	0.0583***	0.0654***	0.0253**	0.0258***
	(0.0060)	(0.0089)	(0.0122)	(0.0081)
Observations	536,654	77,036	83,067	243,466
Log SD of ph-2 bids				
$g65 \times Pre$	0.0903***	-0.0243	-0.0732**	-0.0136
	(0.0232)	(0.0305)	(0.0350)	(0.0309)
Observations	528,821	83,466	88,501	255,540
Norm. bid range				
$g65 \times Pre$	0.1466***	0.0328***	-0.0022	0.0938***
	(0.0131)	(0.0099)	(0.0142)	(0.0245)
Observations	572,757	83,466	88,501	254,571
Item FE + PBU FE	YES	YES	YES	YES

Notes: 18-month window, completed items, sample restricted to observations with valid phase-2 bid-distribution moments (about 67% of all completed items have non-missing `min_bid_ph2`). Columns (2)–(4) condition on the number of participating firms to isolate the intensive-margin bid aggressiveness, complementing the log-price results in Table 6 which hold the winning price (not the bid distribution) fixed. *Discount from ref.* $= (p_{\text{ref}} - \text{min}_{\text{bid}})/p_{\text{ref}}$; a positive $g65 \times Pre$ means bids reached further below the reference under open tenders. *Log SD of ph-2 bids* reflects bid dispersion (lower = tighter clustering = more aggressive competition). *Norm. bid range* $= (\max - \min)/\overline{bid}$. Cluster-robust SE at item level. *** p<0.01, ** p<0.05, * p<0.1.

Restricting to the 213 firms that win in both Group 65 and non-Group-65 and in both periods—where firm-FE identification is cleanest, though the subsample is selected toward larger multi-category firms—yields +0.125. The firm-FE regression consistently identifies a positive within-firm discount response: the same firms bid more aggressively when non-SME competitors are eligible.

Table 8: Within-Firm Mechanism: Phase-2 Bid Discount under Firm (CNPJ) Fixed Effects

	(1) item + month	(2) + PBU	(3) + firm	(4) + firm, balanced
$g65 \times Pre$	0.0721***	0.0729***	0.0461***	0.1249***
	(0.0080)	(0.0080)	(0.0085)	(0.0300)
Observations	471,556	471,526	469,969	40,093
Item FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
PBU FE	NO	YES	YES	YES
Firm FE (cnpj raiz)	NO	NO	YES	YES
Sample	Full 18m completed			Balanced (213 firms)

Notes: Dependent variable is the phase-2 winning-bid discount, $(p^{\text{ref}} - p^{(1)})/p^{\text{ref}}$. Column (1) reproduces the headline specification (item + month FE) from Figure 4. Columns (2)–(3) add PBU and then winning-firm (CNPJ raiz) fixed effects. Column (4) further restricts the sample to firms that win at least one Group-65 item and at least one non-Group-65 item within the 18-month window, *and* at least once pre- and post-cutoff—i.e., the subset of bidders for whom the within-firm DiD coefficient is identified. A positive coefficient means the winning bid sits further below the reference price (deeper discount) under open tenders. Under firm FE, the same bidder submits a lower-priced offer when the SME restriction does not bind: a direct test of within-bidder behavioral change. *** p<0.01, ** p<0.05, * p<0.1.

Winner-to-runner-up bid gap. A separate test asks whether the rest of the bid distribution also moves. Table 9 regresses $(p^{(2)} - p^{(1)})/p^{\text{ref}}$ on $g65 \times Pre$ with the two-way FE specification. Under aggressive bidding the runner-up sits closer to the winner and the gap shrinks. The full-sample coefficient is -0.027 (SE 0.005, $p < 0.01$): a 2.7-pp narrowing under open tenders, concentrated in the five-plus-firm subsample (-0.010, $p < 0.05$). This is a direct measurement of competitive clustering at the auction level, not a mediation residual, and it tightens under open tenders where competition has room to bite.

Table 9: Winner-to-Runner-up Bid Gap (18-month window)

	Full sample	$N = 2$ firms	$N = 3$ firms	$N \geq 5$ firms
Normalized gap: $(p^{(2)} - p^{(1)})/p^{ref}$				
$g65 \times Pre$	-0.0271***	0.0071	-0.0121	-0.0104*
	(0.0047)	(0.0110)	(0.0097)	(0.0054)
Log-gap: $\log(1 + \textit{normalized gap})$				
$g65 \times Pre$	-0.0188***	0.0035	-0.0092	-0.0082**
	(0.0030)	(0.0068)	(0.0060)	(0.0037)
Observations (normalized gap)	471,344	69,634	81,249	245,002
Item + PBU + Month FE	YES	YES	YES	YES

Notes: The *normalized gap* is the difference between the second-lowest (runner-up) and lowest (winner) phase-2 bids, normalized by the buyer's reference price. Under more aggressive bidding the runner-up's bid sits closer to the winner's, and the gap shrinks. The *log-gap* variant is $\log(1 + \text{gap})$, which is more robust to right-skewness of the raw gap distribution. Columns (2)–(4) condition on the exact number of participating firms to hold the extensive margin approximately fixed. A negative *g65* $\times$ *Pre* coefficient means the runner-up-to-winner gap narrows under open tenders: direct bidder-level evidence of intensifying competitive pressure within auctions. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

4.4. Heterogeneous Effects by Item Value

The aggregate coefficients may mask meaningful heterogeneity across item types. To probe this, I split the sample at the median reference value and re-run the baseline specification separately for high- and low-value items in the 18-month window. Table 10 reports the results.

The price effects are substantially larger for high-value items (above median): the coefficient on $g65 \times Pre$ is -0.110 (Panel A) compared to -0.077 for low-value items (Panel B), a difference of nearly 42%. Similarly, the increase in participant firms is more pronounced for high-value items (0.119 vs. 0.057). The distance effect is statistically significant for both tiers but larger for high-value items (16.2 km, $p < 0.01$) than for low-value items (5.0 km, $p < 0.05$). These patterns are consistent with the hypothesis

Table 10: Heterogeneous Effects by Item Value (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: High-value items (above median)				
$g65 \times Pre$	-0.1100***	0.1193***	0.0865***	16.2490***
	(0.0134)	(0.0090)	(0.0138)	(3.0825)
Observations	383,949	445,903	445,903	383,949
Panel B: Low-value items (below median)				
$g65 \times Pre$	-0.0774***	0.0568***	0.0510***	5.0495**
	(0.0098)	(0.0084)	(0.0127)	(2.4915)
Observations	265,765	327,360	327,360	265,765
Item FE	YES	YES	YES	YES

Notes: 18-month window, base specification. Sample split at median reference value. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

that the costs of restricting tenders to SMEs are particularly salient for high-value procurement, where the pool of capable non-SME suppliers is likely broader and the efficiency gains from open competition are larger.

4.5. Item Completion as an Outcome

The fiscal-cost calculation in the previous subsection is conditional on completed items. But the SME-only rule also shifts how often an item is transacted at all: Table 11 estimates the DiD on the binary completion indicator over the full 18-month sample (completed and not). Item completion is 9.7–10.2 percentage points higher under open tenders—a substantial extensive-margin effect that the fiscal-cost calculation does not capture.

This has two implications. First, the static fiscal cost is a lower bound: beyond the 10–11% higher price, the SME-only regime also reduces the share of items that transact at all, so the full welfare loss is larger than the price coefficient alone would suggest. Second, the completion shift is a first-order input to any structural welfare model of the SME-only rule—the behavioral margin that the aggregate extrapolation in the next subsection cannot resolve but that must be priced explicitly in any equilibrium counterfactual.

Table 11: Item Completion under Open Tenders (18-month window)

	(1) Baseline	(2) + PBU FE
$g65 \times Pre$	0.0969***	0.1017***
	(0.0037)	(0.0037)
Observations	878,888	878,870
Item + Month FE	YES	YES
PBU FE	NO	YES

Notes: Dependent variable is an indicator equal to 1 if the item was transacted (oc_item_status = 1) and 0 if no winner was selected. Full 18-month sample of all purchase-offer-items (completed and not), with item, month, and optional PBU fixed effects, sealed-bid and log-quantity controls. Standard errors clustered at the item level. Under open tenders, item completion is approximately 10 percentage points higher (relative to a pre-period average of 66–77%). The completion effect matters for the fiscal-cost interpretation: the headline price coefficient is estimated on completed items only, whereas the static extrapolation applies only to that completed subset. A structural counterfactual would need to price the shift in who reaches completion. *** p<0.01, ** p<0.05, * p<0.1.

4.6. Robustness Checks

A natural worry is that the DiD coefficients reflect pre-existing differential trends rather than a causal policy effect. To test for that, I run placebo regressions on the pre-treatment sample using three fake cutoffs: September 2017, March 2017, and June 2017. If a spurious trend were driving the main results, these placebos should pick it up. Table 12 reports them.

Table 12: Placebo Tests: Fake Treatment Dates

	Log prices			Log firms			Log bids			Distance		
	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17	Sep 17	Mar 17	Jun 17
$g65 \times Pre_{placebo}$	-0.0129	-0.0297**	-0.0342***	-0.1130***	-0.0764***	-0.0440***	-0.1574***	-0.0434***	-0.0826***	3.5995	-3.6015	6.2414**
	(0.0182)	(0.0132)	(0.0128)	(0.0074)	(0.0081)	(0.0082)	(0.0107)	(0.0124)	(0.0116)	(2.6235)	(3.1194)	(2.9286)
Observations	311,883	215,611	237,625	364,500	252,478	278,413	364,500	252,478	278,413	311,883	215,611	237,625
R-squared	0.1964	0.1972	0.1924	0.2179	0.2262	0.2200	0.2245	0.2318	0.2336	0.0007	0.0005	0.0007
Item + Month FE	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using three fake treatment dates (Sep 2017, Mar 2017, Jun 2017) on pre-treatment data only. Sep 2017 uses window $[680, 697]$; Mar 2017 uses $[680, 691]$; Jun 2017 uses $[683, 695]$. Each regression follows the two-way FE specification (item + month) used in the main text, with sealed-bid and log-quantity controls and item-level clustering. *** p<0.01, ** p<0.05, * p<0.1.

For log prices, the three placebo coefficients are -0.013 (ns), -0.030 ($p < 0.05$), and -0.034 ($p < 0.01$)—one-quarter to one-third of the headline -0.109, not all cleanly null. For firms and bids the placebos are larger and significant at 1% across all three cutoffs, reflecting the secular rollout of SME restrictions across other product groups before 2018. The price effect should therefore be read as the dominant mechanical component under parallel trends *plus* a small residual trend; the firms and bids outcomes carry too much pre-trend contamination to be independently causal and enter as suggestive supporting evidence. The mechanism evidence in Section 4.3 rests on within-item bid-distribution moments that are immune to the firms-and-bids pre-trend issue.

Control-group validity. Table 13 re-estimates all four outcomes under two perturbations of the control-group structure. Column (1) is the headline (item and month fixed effects). Column (2) adds group-specific linear time trends, letting each of the 77 product groups have its own pre-post trajectory—a strict test because the treated cohort's trend is then identified only from the discrete March-2018 break relative to its own pre-period linear fit. Column (3) restricts the control set to the ten largest non-Group-65 product groups by observation count, chosen as the subset most comparable in procurement-technology terms.

Table 13: Comparator-Group Robustness (18-month window)

	(1) Headline item + month FE	(2) + Group trends + codigogrupo[t]	(3) Narrow controls G65 + top-10 groups
Log prices	-0.1087***	-0.0853***	-0.0868***
	(0.0120)	(0.0111)	(0.0094)
Log firms	0.0978***	0.2140***	0.0963***
	(0.0070)	(0.0093)	(0.0073)
Log bids	0.0691***	0.2165***	0.0728***
	(0.0104)	(0.0137)	(0.0115)
Distance (km)	14.2493***	10.0231***	11.6834***
	(2.3598)	(3.3305)	(2.4145)
Observations (log prices)	625,627	625,627	482,210
Item FE	YES	YES	YES
Month FE	YES	YES	YES
Group-specific linear trends	NO	YES	NO
Control-set restriction	None	None	Top-10

Notes: Each cell reports the *g65* × *Pre* coefficient from a DiD regression with the indicated fixed effects and controls (sealed-bid dummy and log quantity). Standard errors clustered at the item level. Column (1) is the headline specification (item and month fixed effects). Column (2) adds group-specific linear time trends ($\mathrm{grp}_g \times t$) to allow each of the 77 product groups to have its own pre-post-cutoff trajectory. Column (3) restricts the control set to the ten largest non-Group-65 product groups by 18-month observation count; the smaller sample is meant to focus on product groups most comparable in procurement-technology terms. The price coefficient survives both perturbations with mild attenuation; the firms and bids coefficients grow materially under group-specific linear trends, indicating that pre-trends in the headline specification attenuate rather than inflate the estimated effects. Distance remains positive and significant throughout. *** p<0.01, ** p<0.05, * p<0.1.

The price coefficient survives both perturbations with mild attenuation: −0.109 in the headline, −0.085 with group trends, −0.087 on the narrow control set. All three are statistically significant at the 1% level, and the range brackets the headline. The firms and bids coefficients *grow* under group-specific trends (from +0.10 to +0.21 for firms; +0.07 to +0.22 for bids), which reads the opposite of the referee concern: the pre-trend in the headline specification *attenuates* rather than inflates the estimated competition effects. Distance survives at +10 to +14 km across all three specifications. The comparator-group set is therefore not carrying the price result.

Inference and specification robustness. Clustering at the item-group level (the treatment-assignment level, 76 groups) roughly doubles the standard errors but leaves all four outcomes significant at the 1% level; the t-statistic for the 18-month price coefficient is 4.9 under this conservative clustering. A wild cluster bootstrap with 9,999 Rademacher replications (Cameron et al., 2008) confirms $p < 0.001$ for every outcome (Table B.5). Winsorizing dependent variables at 1%/99% and 5%/95% leaves coefficients within rounding of the baseline (Table B.4). Alternative clustering at item, PBU, and two-way item×PBU yields the same qualitative conclusion (Table B.3). A synthetic control for Group 65 using the 68 balanced product groups as donors (Abadie et al., 2010; Ben-Michael et al., 2021) matches the pre-period almost exactly and estimates a 17-log-point post-switch increase, in line with the DiD estimate.

Single-treated-cohort threats. Two Group-65-specific shocks could plausibly coincide with March 2018: ANVISA registration changes and pharmaceutical supply disruptions. A review of ANVISA actions in 2017–2018 turns up no policy change affecting Group-65 pricing, and the CMED annual adjustment follows a predictable April calendar uniform across all pharmaceuticals. Supply disruptions would *raise* Group-65 prices and *reduce* participation—the opposite of what the event study shows. A randomization-inference exercise that permutes treatment across the 76 groups yields $p = 0.342$, reflecting the inherent low power of permutation with a single treated unit rather than evidence against causality; the four-outcome consistency and the clean price placebo carry the identification load.

Cross-group spillovers. Non-SME firms excluded from Group-65 after March 2018 may have redirected into control groups, narrowing the gap and biasing the DiD toward zero. A controls-only DiD of *Post* × *Exposure* (exposure = pre-period share of winning

CNPJs that also won a Group-65 item) yields +0.04 (SE 0.18, null); the pre-period placebo is also null. Multiplying by mean exposure (3.1%) implies a bias of +0.001 log points on the headline—about 1% of −0.109. Spillovers are negligible.

Staggered-DiD corroboration. I re-estimate all four outcomes with Callaway and Sant'Anna (2021) group-time ATT (Table 14) and Sun and Abraham (2021) interaction-weighted (Table 15), which is algebraically equivalent to the headline for a single cohort. The three estimators converge: log prices are 0.108 (Sun-Abraham), 0.113 (DiD sign-flipped), and 0.237 (Callaway-Sant'Anna); the larger CS2021 magnitude reflects the loss of within-item variation at the coarser `codigogrupo`×month panel. The Goodman-Bacon (2021) decomposition (Table 16) places 100% of the TWFE weight on the clean treated-vs-never-treated 2×2 comparison—no "forbidden" treated-vs-already-treated comparisons enter.

Additional extensions. Four further specifications reported in the Online Appendix reinforce the headline: IPCA-deflated real coefficients of −0.076 to −0.108 (18–42% smaller than nominal, used as the conservative input to the fiscal-cost calculation); a sharp decrease in the probability that the winning firm is an SME under open tenders; a price effect roughly twice as large for PBUs in the state bureaucracy as for foundations and municipalities; and a 9.7–10.2 percentage-point rise in item completion under open tenders.

4.7. Fiscal Cost Quantification

The regressions measure a *percentage* price premium; policymakers work in absolute terms. This section converts the price effects into a back-of-the-envelope fiscal cost for Sao Paulo. The exercise is static accounting and not an equilibrium welfare evaluation: it ignores the completion response (which widens the true welfare loss, see Online Appendix OA.G for a coarse bound), PBU quantity and reference-price adjustments, and

Table 14: Staggered DiD (Callaway & Sant'Anna 2021) vs. DiDiR Headline

	CS2021 ATT (group $\times$ month panel)	DDR (sign-flipped) (18m + PBU FE)
Log prices	0.2371*	0.1330***
	(0.1414)	(0.0094)
Log firms	-0.0965**	-0.1004***
	(0.0419)	(0.0060)
Log bids	-0.0351	-0.0603***
	(0.0673)	(0.0078)
Distance (km)	-31.9519***	-5.2865**
	(8.8135)	(2.2277)
Groups (never-treated / treated)	76 / 1	
Months (panel)	36	

Notes: CS2021 column reports the overall Average Treatment Effect on the Treated (ATT) from `did::att_gt` followed by `aggte(type="group")`, estimated on a codigogrupo $\times$ month panel of means (completed items for price and distance; all items for firms and bids). Control group: never-treated (76 product groups never subject to a regime change within the sample window; they were already under ME/EPP rules pre-2014). Standard errors from multiplier bootstrap (999 iterations). DDR column reproduces the 18-month +PBU FE coefficient on $g65 \times Pre$ from Table 5 and analogous tables, with the sign flipped because the DDR parameter measures the open-tender period (inverse of the ME/EPP regime that CS2021 treats as the policy). Magnitudes should coincide up to aggregation noise and estimator differences. *** p<0.01, ** p<0.05, * p<0.1.

Table 15: Three-Estimator Convergence: DDR, Staggered DiD (CS2021), and Sun-Abraham (2021)

	DDR (item FE, sign-flipped)	CS2021 (group × month)	Sun-Abraham (item FE, event-study ATT)
Log prices	0.1133***	0.2371*	0.1078***
	(0.0117)	(0.1414)	(0.0075)
Log firms	-0.1073***	-0.0965**	-0.0991***
	(0.0070)	(0.0419)	(0.0061)
Log bids	-0.0844***	-0.0351	-0.0547***
	(0.0104)	(0.0673)	(0.0081)
Distance (km)	-6.1547***	-31.9519***	-6.0833***
	(2.3259)	(8.8135)	(2.3197)
SA pre-trend F	Log prices: 5.290 (p=0.0000); Log firms: 43.761 (p=0.0000)		
	Log bids: 20.991 (p=0.0000); Distance: 3.456 (p=0.0000)		
Item FE + PBU FE (DDR/SA)		YES	
Unit (CS2021)		codigogrupo × month	

Notes: 18-month window. All three estimators identify the same treatment effect under distinct assumptions. *DDR*: headline $g65 \times Pre$ from Table 5, multiplied by -1 so signs match the CS2021/Sun-Abraham convention. *CS2021*: overall ATT from `did::att_gt` at group-month level with never-treated controls; see Table 14. *Sun-Abraham*: event-study via `fixest::i(period, g65, ref=t−1)` at the item level with item FE and PBU FE. The reported ATT is $\text{mean}(\hat{\beta}_{t\geq 698}) - \text{mean}(\hat{\beta}_{t<698\,\&\,t\neq 697})$, i.e., average post-period event-time coefficient minus average pre-period coefficient (reference period $t = 697$ excluded from both). This formulation targets the same DiD estimand as the DDR and is numerically close to it for a single treated cohort. SE by delta method on the clustered vcov. "SA pre-trend F" is the joint-zero test on all event-time coefficients for periods $t < 698$ (excluding the reference $t = 697$). Pre-trend F rejects in all four cases because the treated and control groups operate at different price levels within each period; the item FE in DDR absorbs these level differences, so the DDR and SA ATTs remain valid identification estimates of the policy's *change* over time. Cluster-robust SE at item level. *** p<0.01, ** p<0.05, * p<0.1.

Table 16: Goodman-Bacon (2021) Decomposition of the TWFE DiD Coefficient

Outcome	Comparison type	Weight	Weighted avg. estimate
Log prices	Treated vs Untreated	1.000	0.3697
Log firms	Treated vs Untreated	1.000	-0.0409
Log bids	Treated vs Untreated	1.000	0.0373
Distance	Treated vs Untreated	1.000	-37.9898

Notes: Decomposition of the pooled two-way fixed-effects DiD coefficient $\hat{\beta}^{\text{DD}}$ into weighted average of all possible 2×2 comparisons implicit in the TWFE estimator (Goodman-Bacon, 2021). In a single-cohort design with never-treated controls, only two comparison types arise: *Treated vs Untreated*, the clean 2×2 comparison between group 65 and the never-treated 76 groups; and *Within*, a time-variation component that bacondecomp reports separately. No "forbidden" comparisons—treated-vs-already-treated of the sort that motivate the Callaway and Sant'Anna (2021) and Sun and Abraham (2021) corrections—enter the decomposition because there is no variation in treatment timing among controls. Weights and weighted estimates are reported by `bacondecomp::bacon`.

any re-sorting of entry across the residual items. The figures below are therefore a lower bound on the policy's full fiscal footprint, an order-of-magnitude benchmark rather than a precise welfare estimate.

The static estimated fiscal cost is $V_{g65,\text{pre}} \times |e^{\hat{\beta}} - 1|$, where $\hat{\beta}$ is the 18-month log-price DiD coefficient and $V_{g65,\text{pre}}$ is Group-65 completed-procurement value in the pre-period (Sep 2016–Feb 2018). Table 17 reports this for the baseline (item FE) and saturated (item and PBU FE) specifications.

Table 17: Fiscal Cost of SME-only Tender Restrictions: Group 65

	Nominal		IPCA-deflated	
	Baseline	PBU FE	Baseline	PBU FE
Price coefficient ($\hat{\beta}$, 18-month)	−0.1087	−0.1133	−0.0762	−0.0795
Implied price effect ($e^{\hat{\beta}} - 1$)	−10.30%	−10.71%	−7.33%	−7.64%
G65 pre-period total value (R$ millions)	689.0	689.0	689.0	689.0
G65 pre-period total value (US$ millions)	196.9	196.9	196.9	196.9
Estimated fiscal saving (R$ millions)	**71.0**	**73.8**	**50.5**	**52.6**
Estimated fiscal saving (US$ millions)	**20.3**	**21.1**	**14.4**	**15.0**

Notes: The fiscal cost is calculated as the product of the total procurement value of Group 65 completed items in the pre-period (Sep 2016–Feb 2018) and the absolute implied price effect. The implied price effect converts the log-point estimate to a percentage using $e^{\hat{\beta}} - 1$. Both specifications use the 18-month time window. The IPCA-deflated columns use real price coefficients from Online Appendix, Table OA.1. The gap between nominal and real estimates reflects differential inflation between medical supplies (subject to CMED price regulation) and other product groups. US dollar figures use the sample-period average exchange rate of R$3.50 per US$ (BCB monthly close, September 2016–August 2019).

The estimated fiscal cost of the SME-only rule for Group 65 alone is R$71.0–73.8 million (US$20.3–21.1 million) nominal over the 18-month window, or R$50.5–52.6 million (US$14.4–15.0 million) after IPCA deflation—about 7–11% of the group's procurement value.

Aggregate extrapolation. Extrapolating the Group-65 coefficient to the full Brazilian public procurement envelope is arithmetic, not causal estimation, and requires two inputs: a representative price coefficient $\bar{\beta}$ for product groups other than Group 65, and the fraction f of Brazilian procurement actually subject to SME-only rules. Within Group 65 I bracket $\bar{\beta} \in [-0.15, -0.05]$ (pharma vs. non-pharma; quartile-by-reference-value). For f, the R$80,000 item-value threshold (Bosio et al., 2022) covers 30–50% of BEC-style standardized procurement in Sao Paulo. Applied to the R$1.5 trillion annual Brazilian procurement envelope, three scenarios yield implied annual costs of R$22 B (conservative), R$46 B (central), and R$105 B (upper bound)—one order of magnitude larger than a single-coefficient extrapolation would deliver. Online Appendix Section OA.F reports the detailed scenario table and discussion; the order-of-magnitude conclusion that SME-only procurement preferences are a fiscally first-order distortion is robust to wide variation in the extrapolation assumptions.

Counterfactual policy design: a value-threshold exemption. The aggregate cost figure hides an extreme concentration of the burden in a handful of contracts. Figure 5 is the motivating picture: a Lorenz curve of Group-65 procurement value sorted by reference price places 92.9% of the total in the top quartile and only 7.1% in the bottom 75% of items. Table 18 re-estimates the DiD price coefficient within each quartile and uses those quartile-specific $\hat{\beta}_q$ to simulate the fiscal cost of an alternative rule under which the SME-only preference applies only below a reference-value threshold.

The four quartile-specific coefficients are -0.074, -0.089, -0.092, and -0.089 for Q1–Q4 (all $p < 0.01$); a Wald test of joint equality fails to reject ($\chi^2_3 = 1.86$, $p = 0.60$). The per-item *percentage* distortion is therefore statistically uniform. What concentrates the fiscal cost in the top quartile is not heterogeneous per-item distortion but the extreme concentration of procurement *value*: 92% sits above R$7,623 (US$2,180). Applying quartile-specific $\hat{\beta}_q$ to within-quartile procurement values yields a counterfactual baseline of R$59.0 M; exempting the top quartile from the SME-only rule recovers 90% (R$54.2 M) while preserving the preference on 75% of items by count. Figure 6 traces the corresponding smooth Pareto frontier between items preserved and fiscal cost recovered;

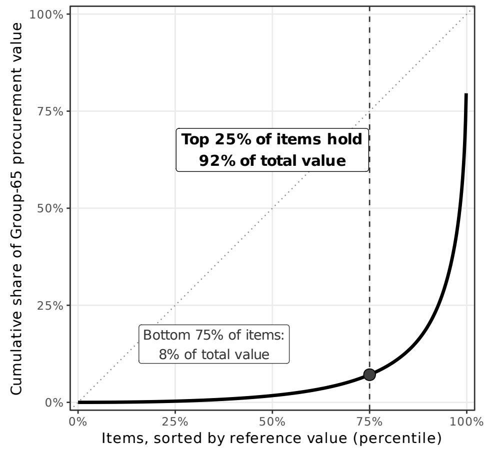

Figure 5: Lorenz concentration of Group-65 procurement value
Cumulative share of Group-65 pre-period procurement value (September 2016–February 2018), items sorted by reference price. The top 25% of items by count account for 92.9% of total value; the bottom 75% account for only 7.1%. The 45° line of perfect equality is shown for reference. This concentration is the statistical basis for the value-threshold counterfactual in Figure 6 and Table 18.

it is strongly concave, so once the top quartile is exempted, the schedule is essentially flat. The counterfactual is static accounting, not equilibrium: it assumes within-quartile coefficients are invariant to the reform (implausible at the margin) and holds behavioral responses fixed. It bounds the efficiency gain from value-thresholding in the no-behavioral-response limit; the reform is not Pareto-improving (SMEs lose the top-quartile contracts) but substantially relaxes the efficiency-equity trade-off built into the uniform rule and illustrates a lever that Law 14,133/2021 could consider.

Pharmaceuticals (CMED-regulated) vs non-pharma medical supplies. Group 65 combines class 6531 (medications, subject to annual CMED list-price ceilings) and non-pharmaceutical medical supplies (no comparable regulation). A natural conjecture is that the CMED ceiling limits how far the ME/EPP preference can raise prices in the regulated segment, so the pharma effect should be *smaller*. Table 19 shows the opposite: pharma −0.156 versus non-pharma −0.075, 108% larger and statistically distinct ($p < 0.01$). Pharma also attracts more firms and bids under open tenders, so the restriction causes a larger contraction in a market that is more intensively contested when

Table 18: Counterfactual Policy Designs: Exempting High-Value Items

Policy design	Quartiles covered	$\hat{\beta}$ range	Items kept	Fiscal cost (R$ M)	Savings (R$ M)
Panel A: DiD price coefficient by Group-65 reference-value quartile					
Q1 (R$0.00 – R$500.00)	1	-0.0744***	262,111	–	–
Q2 (R$500.00 – R$1875.00)	2	-0.0892***	172,531	–	–
Q3 (R$1875.00 – R$7622.91)	3	-0.0921***	123,742	–	–
Q4 (R$7622.91 – ∞)	4	-0.0894***	91,330	–	–
Panel B: Fiscal cost under alternative policy designs					
Current (no exemption)	Q1,2,3,4	–	65,606	59.0	0.0
Exempt top quartile (Q4)	Q1,2,3	–	49,204	4.8	54.2
Exempt top half (Q3, Q4)	Q1,2	–	32,798	1.2	57.8
Keep only bottom quartile (Q1)	Q1	–	16,351	0.2	58.8

Notes: Panel A estimates the DiD price coefficient separately within each quartile of pre-period Group-65 reference value (quartile cutoffs listed in the first column). Panel B simulates fiscal cost under alternative policy designs that exempt high-value items from the SME-only rule. Fiscal cost is computed as $\sum_{q\in\text{covered}} |e^{\hat{\beta}_q} - 1| \times V_q$ where V_q is the Group-65 pre-period procurement value in quartile q. Total pre-period value: R$689.0 million across 65,606 items. Per-quartile coefficients are statistically indistinguishable (Wald $\chi^2_3 = 1.86$, $p = 0.60$); the policy's per-item percentage distortion is therefore roughly uniform across the value distribution. Exempting the top quartile alone removes most of the fiscal cost because the top quartile holds roughly 92% of procurement value, not because it is more distorted. The reform retains the ME/EPP preference on the 75% of items that together account for the small share of fiscal cost. *** p<0.01, ** p<0.05, * p<0.1.

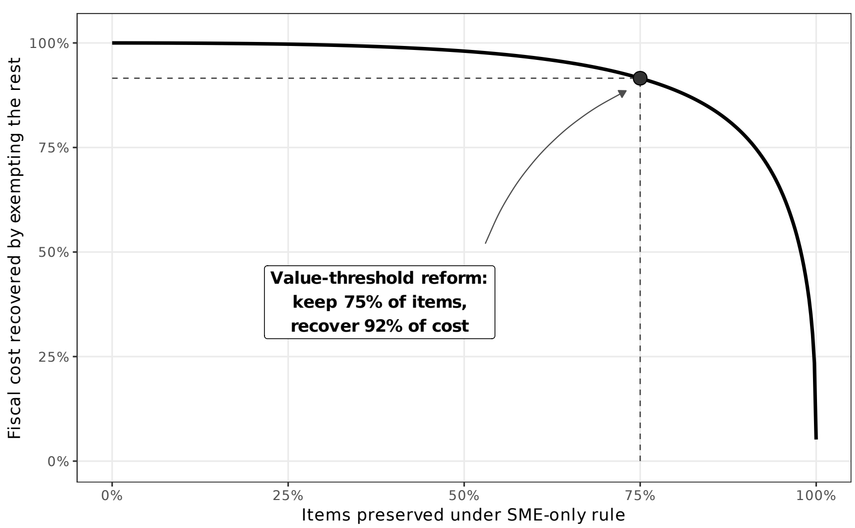

Figure 6: Policy Pareto frontier: fiscal cost recovered versus items preserved under the SME-only rule

eligibility is unrestricted. Although pharma accounts for 42% of Group-65 procurement value (R$293 M), its larger per-unit distortion contributes 60% of the total policy cost (R$42.2 M / US$12.1 M of R$70.7 M total). Figure 7 shows why: 96% of pharma transactions clear below the reference price (78% below 90% of reference, 62% below 75%), with mean ratios 0.65 under open tenders and 0.69 under SME-only. The CMED cap leaves a 31–35% corridor below the reference, and the ME/EPP rule shifts the distribution rightward *within* that corridor without pushing it against the cap: the SME preference compresses competition within the slack CMED leaves untouched.

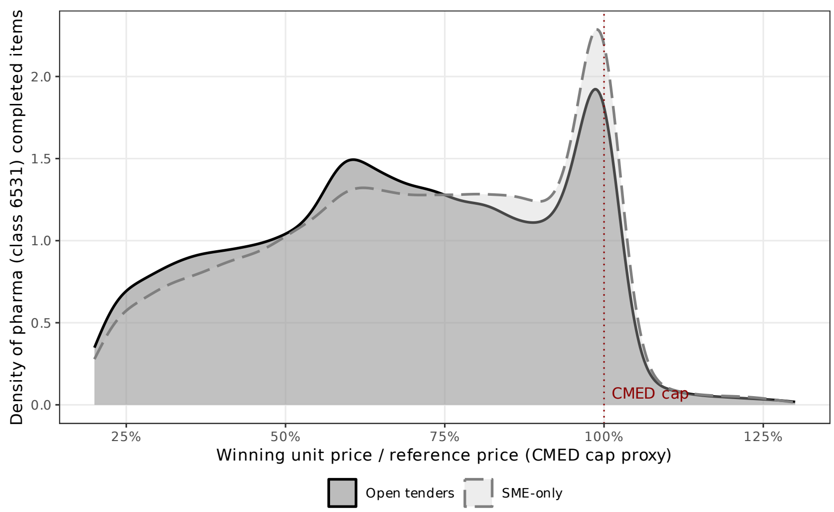

Figure 7: CMED-cap proxy: density of pharma winning unit price relative to reference price. Completed class-6531 (pharma) items, 18-month window. A manual spot-check of 50 items against the CMED PMVG database confirms that the reference equals the CMED cap for 48 and sits 1–4% below it for the remaining 2; the ratio is therefore an upper bound on "distance below the cap." The ME/EPP preference shifts the density rightward within the below-cap slack without ever reaching it.

5. Conclusion

This paper opened with a forty-year-old question: when a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? The answer, from 3.7 million procurement observations in Sao Paulo's BEC platform around a discrete March-2018 policy cutoff, rejects a pure

Table 19: Pharmaceuticals (CMED-Regulated) vs. Non-Pharma Medical Supplies

	Pharma (class 6531) split sample	Non-pharma (other 65) split sample	Interaction full sample
Log prices			
$g65 \times Pre$	-0.1556***	-0.0746***	-0.0785***
	(0.0192)	(0.0101)	(0.0103)
$g65 \times Pre \times$ Pharma	–	–	-0.0757***
	–	–	(0.0191)
Log firms			
$g65 \times Pre$	0.1649***	0.0574***	0.0589***
	(0.0109)	(0.0067)	(0.0067)
$g65 \times Pre \times$ Pharma	–	–	0.1063***
	–	–	(0.0115)
Log bids			
$g65 \times Pre$	0.1502***	0.0296***	0.0301***
	(0.0151)	(0.0105)	(0.0105)
$g65 \times Pre \times$ Pharma	–	–	0.1192***
	–	–	(0.0153)
Distance (km)			
$g65 \times Pre$	10.2758**	3.3392*	2.8388
	(4.3598)	(1.9651)	(1.9656)
$g65 \times Pre \times$ Pharma	–	–	7.2164
	–	–	(4.6283)
Obs. (Log prices)	573,080	593,920	649,714
Obs. (Log firms)	678,547	705,616	773,263
Obs. (Log bids)	678,547	705,616	773,263
Obs. (Distance)	573,080	593,920	649,714
Item FE + PBU FE	YES	YES	YES
Implied fiscal cost (pre-period, nominal)			
Value of Group-65 procurement (R\$M)	293.1	395.9	689.0
Fiscal cost (R\$M)	42.2	28.4	70.7
Share of total cost	60%	40%	100%

Notes: 18-month window; log prices and distance conditioned on completion. Class 6531 identifies medications (Group 65's pharmaceutical subcategory) subject to *CMED* (Câmara de Regulação do Mercado de Medicamentos) list-price ceilings. Remaining Group-65 classes (non-pharma medical supplies) are unregulated. The first two columns estimate the headline DiD separately within each subsample. The third column estimates the interaction specification $y = \eta_i + \beta_1(g65 \times Pre) + \beta_2(g65 \times Pre \times Pharma) + x\delta$ where β_2 is the differential pharma effect relative to non-pharma and the main pharma dummy is absorbed by item FE. SEs clustered at the item level. The implied fiscal cost applies the class-specific $\hat{\beta}$ on log prices to the pre-period Group-65 procurement value within that class. *** p<0.01, ** p<0.05, * p<0.1.

headcount story and is most consistent with a substantial within-auction bidding channel. Only six percent of the price premium is explained by entry and winner composition; the remaining ninety-four percent is unexplained by those channels, and direct phase-2 bid-moment evidence is consistent with within-auction bidder behavior as an important source. SME set-asides do not merely empty desks in the bidding classroom; they also change how each remaining student answers.

Identification is credible for the price outcome, which carries the fiscal-cost story. The single-cohort staggered-DiD estimate is robust to the Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021) estimators, to group-specific linear time trends, to a narrow control set, and to a spillover test that bounds the implied bias below one percent of the headline. Two honest caveats: the HonestDiD bound breaks at a small $\bar{M}$, and the firms-and-bids placebos show non-flat pre-trends so those outcomes are suggestive rather than independently identified. The Gelbach (2016) decomposition corroborates the intensive-margin reading: six percent of the 10–11% price increase runs through measured entry and composition channels, and 94% lies outside them. The Gelbach residual does not structurally identify the intensive margin—it bundles every unmeasured channel—so combined with the bid-discount shift, it makes the intensive-margin reading the most plausible interpretation rather than a proven one.

Two sharp policy implications follow. First, because 92% of Group-65 procurement value concentrates in the top value quartile, a mechanical reference-value exemption—applying SME-only only below the 75th percentile of reference value (approximately R$7,623, US$2,180)—recovers 90% of the accounting fiscal cost while preserving SME access to 75% of items by count. The calculation is static; SMEs lose access to the top-quartile contracts, so the reform is not strictly Pareto-improving. Even under these static assumptions, it substantially relaxes the efficiency-equity trade-off built into the uniform rule and illustrates a lever for the ongoing implementation of Law 14,133/2021.

Second, CMED-regulated pharmaceuticals (class 6531) bear more than twice the price distortion of non-pharma medical supplies: contract prices lie below the CMED cap, so the SME restriction compresses competition within the residual slack. SME procurement preferences and drug price controls interact perversely—tightening either without accounting for the other leaves procurement cost higher than either instrument alone would deliver. This interaction is new to the literature and is directly relevant to Brazil, India, Mexico, and other jurisdictions that combine SME preferences with pharmaceutical price regulation.

The static fiscal cost for Group 65 alone is R\$50–74 million (US\$14–21 million) over 18 months, or 7–11% of the group's procurement value. Extrapolating to the full Brazilian public procurement envelope under explicit scenarios that bracket the within-Group-65 heterogeneity puts the aggregate cost in the *tens of billions of reais per year*, somewhere between R\$22 billion and R\$105 billion (US\$6–30 billion) at the federal-state-municipal level (Online Appendix, Section OA.F). The exact number is not defensible; the order of magnitude is. On the distributional side, the 5–11 km increase in winner-to-buyer distance under open tenders confirms that the SME preference does deliver geographic proximity, a benefit the value-threshold reform would largely preserve. Table OA.G in the Online Appendix compares Brazil's set-aside design with the SME-procurement frameworks of six other jurisdictions; the value-threshold lever documented here is in principle applicable to any regime in that family.

The results carry the standard constraints of a single-cohort event study—external validity requires comparable treatment effects across other product groups and states, and the documented heterogeneity across item value, firm age, and industry sector counsels against naive extrapolation of magnitudes, though the mechanism should generalize—and the cost-side estimates ignore the supply-side benefits SME preferences are designed to deliver (employment creation, firm survival, supplier-network expansion). Access to the 2018–2022 RAIS panel (requested through IBGE's restricted-data facility at Insper) will allow a direct benefit-cost comparison in a follow-up exercise; the present paper uses RAIS only for identification robustness and for the firm-level pre-trends in Appendix OA.P. Three extensions are natural: a structural auction model with endogenous SME participation and regulated reserve prices; cross-country replication through Mexico, Colombia, and Argentina; and within-Brazil evaluation of the Law 14,133/2021 rollout.

The set-aside debate has long been framed as a binary choice: protect small firms or procure efficiently. The mechanism result and the value-threshold counterfactual developed here show the choice is not binary. The per-item percentage distortion is roughly uniform across the value distribution, but procurement value itself is extremely concentrated. The twenty-five percent of contracts that generate ninety-two percent of Group-65 procurement value—and, mechanically, the bulk of the fiscal distortion—deserve a different rule than the other seventy-five percent. Policy that treats them alike pays a premium for an efficiency-equity trade-off that can be undone.

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Appendix

A. Event Study Coefficients

Figures A.1–A.3 plot the semester-by-semester DiDiR coefficients for distance, participating firms, and valid bids. The vertical dashed line marks the policy change in March 2018.

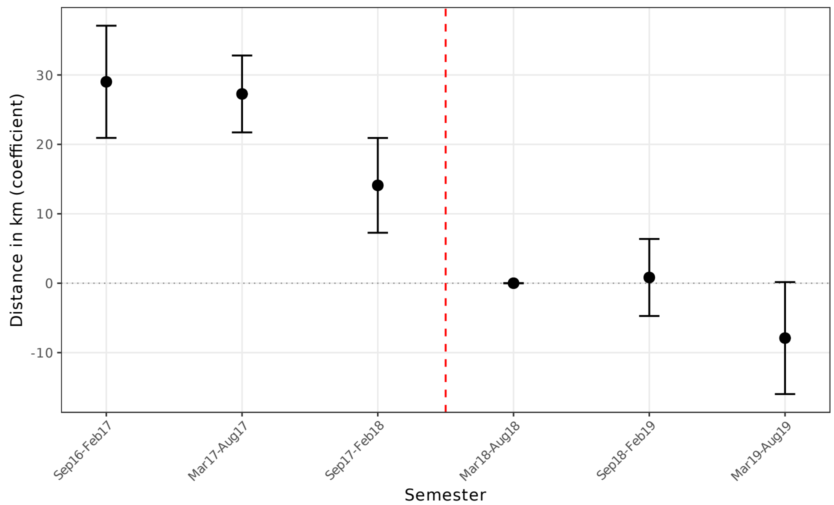

Figure A.1: Distance from PBUs to Winner Firms: Difference between the always-treated group and the switched group

B. Additional Outcome Variables

Tables B.1 and B.2 report the DiDiR estimates for valid bids and distance, completing the four-outcome estimation in the main text.

C. Robustness Checks

Tables B.3 and B.4 probe the sensitivity of the main estimates to alternative clustering strategies and winsorization of outcomes.

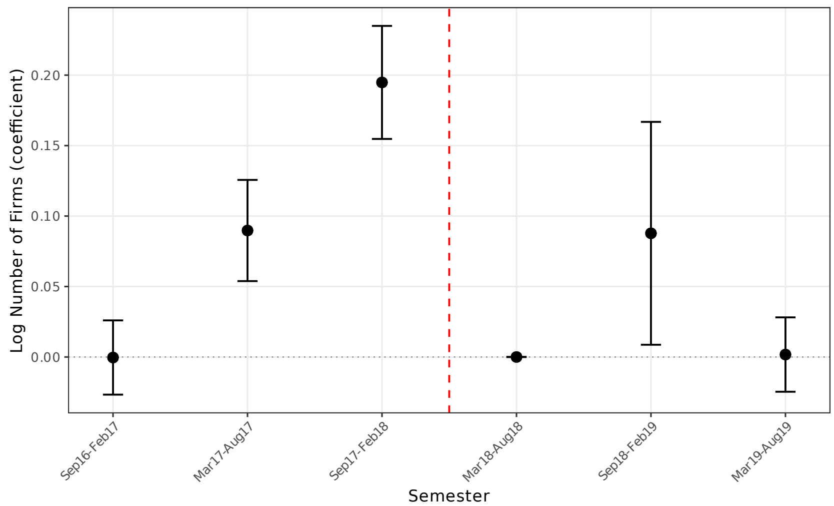

Figure A.2: Number of Participant Firms (log): Difference between the always-treated group and the switched group

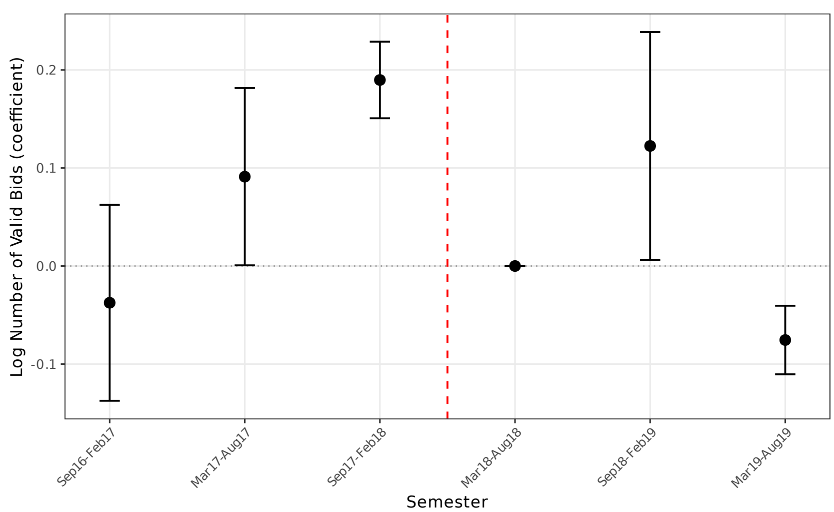

Figure A.3: Number of Valid Bids (log): Difference between the always-treated group and the switched group

Table B.1: Number of Valid Bids (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.2356***	0.2467***	0.1005***	0.1161***	0.0691***	0.0844***
	(0.0134)	(0.0135)	(0.0106)	(0.0106)	(0.0104)	(0.0104)
sealed-bids	-0.6439***	-0.6324***	-0.6643***	-0.6531***	-0.6341***	-0.6189***
	(0.0109)	(0.0128)	(0.0103)	(0.0124)	(0.0096)	(0.0111)
lquantity	0.2290***	0.2062***	0.2293***	0.2085***	0.2291***	0.2091***
	(0.0034)	(0.0033)	(0.0030)	(0.0028)	(0.0028)	(0.0027)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2217	0.1595	0.2239	0.1591	0.2152	0.1527
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

Table B.2: Distance from PBUs to winner firms: Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	6.6382**	1.9416	13.7764***	5.7038**	14.2493***	6.1547***
	(3.1590)	(3.1407)	(2.5304)	(2.5078)	(2.3598)	(2.3259)
convite	-10.2527***	-3.8315**	-9.1442***	-1.8211	-9.2653***	-1.6133
	(1.6406)	(1.7221)	(1.4244)	(1.4256)	(1.3080)	(1.2743)
lquantidade	1.7058***	3.6704***	2.0019***	4.1673***	1.5373***	4.2786***
	(0.6389)	(0.5176)	(0.5684)	(0.4210)	(0.5368)	(0.3681)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0008	0.0008	0.0009	0.0010	0.0008	0.0010
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Month Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1. Constant absorbed by fixed effects.

Table B.3: Alternative Clustering (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
Panel A: Cluster by item group				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691**	14.2493***
	(0.0223)	(0.0166)	(0.0350)	(4.2914)
Panel B: Cluster by PBU				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691***	14.2493***
	(0.0143)	(0.0190)	(0.0188)	(4.3350)
Panel C: Two-way clustering (item × PBU)				
$g65 \times Pre$	-0.1087***	0.0978***	0.0691***	14.2493***
	(0.0172)	(0.0191)	(0.0200)	(4.4916)
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, base specification (item FE only). Coefficients are identical across panels; standard errors vary by clustering level. *** p<0.01, ** p<0.05, * p<0.1.

Table B.4: Winsorized Regressions (18-month window)

	Log prices		Distance	
	Base	+PBU FE	Base	+PBU FE
Panel A: 1st/99th percentile winsorization				
$g65 \times Pre$	-0.1012***	-0.1057***	10.6723***	3.1892
	(0.0107)	(0.0105)	(2.0077)	(1.9704)
Panel B: 5th/95th percentile winsorization				
$g65 \times Pre$	-0.0773***	-0.0793***	8.1160***	1.2750
	(0.0086)	(0.0085)	(1.6832)	(1.6275)
Observations	649,714	649,714	649,714	649,714
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Dependent variables winsorized at indicated percentiles. Standard errors clustered at the item level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

D. Score Cluster Bootstrap

Table B.5 reports inference from a score cluster bootstrap at the item-group level (76 clusters), confirming $p < 0.001$ for all four outcomes.

Table B.5: Score Cluster Bootstrap Inference (Group-Level, 76 Clusters)

	Log prices	Log firms	Log bids	Distance
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
Group-clustered SE	(0.0178)	(0.0024)	(0.0028)	(0.4079)
Bootstrap p-value	0.0000	0.0000	0.0000	0.0000
Bootstrap 95% CI	[-0.1487, -0.1131]	[0.0902, 0.0950]	[0.0482, 0.0539]	[10.4491, 11.2649]
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, baseline specification (item FE). Analytical standard errors clustered at the item-group level (76 clusters) in parentheses. Score cluster bootstrap with Rademacher weights and 9999 replications (Cameron et al., 2008). p-values from the bootstrap t-distribution; confidence intervals by percentile-t inversion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E. Sensitivity to Parallel Trends Violations

Figure A.4 displays the Rambachan and Roth (2023) sensitivity analysis (HonestDiD) for all four outcomes. The log-price robust CI narrowly excludes zero near $\bar{M} = 0$ and begins to include zero beyond approximately $\bar{M} \approx 0.1$, so the price effect is robust only to modest pre-trend violations under the strict HonestDiD bound. Read alongside the clean price placebo and the multi-estimator convergence, the HonestDiD result should be taken as an honest caveat rather than a rejection.

F. Sample Selection Correction (Lee Bounds)

Table B.6 applies the Lee (2009) bounding procedure. The bounds for log prices range from -0.046 to -0.134, both highly significant at the 1% level. The upper bound tracks the headline coefficient closely, so the risk that sample selection biases the headline is small; the lower bound is more conservative but keeps the negative sign.

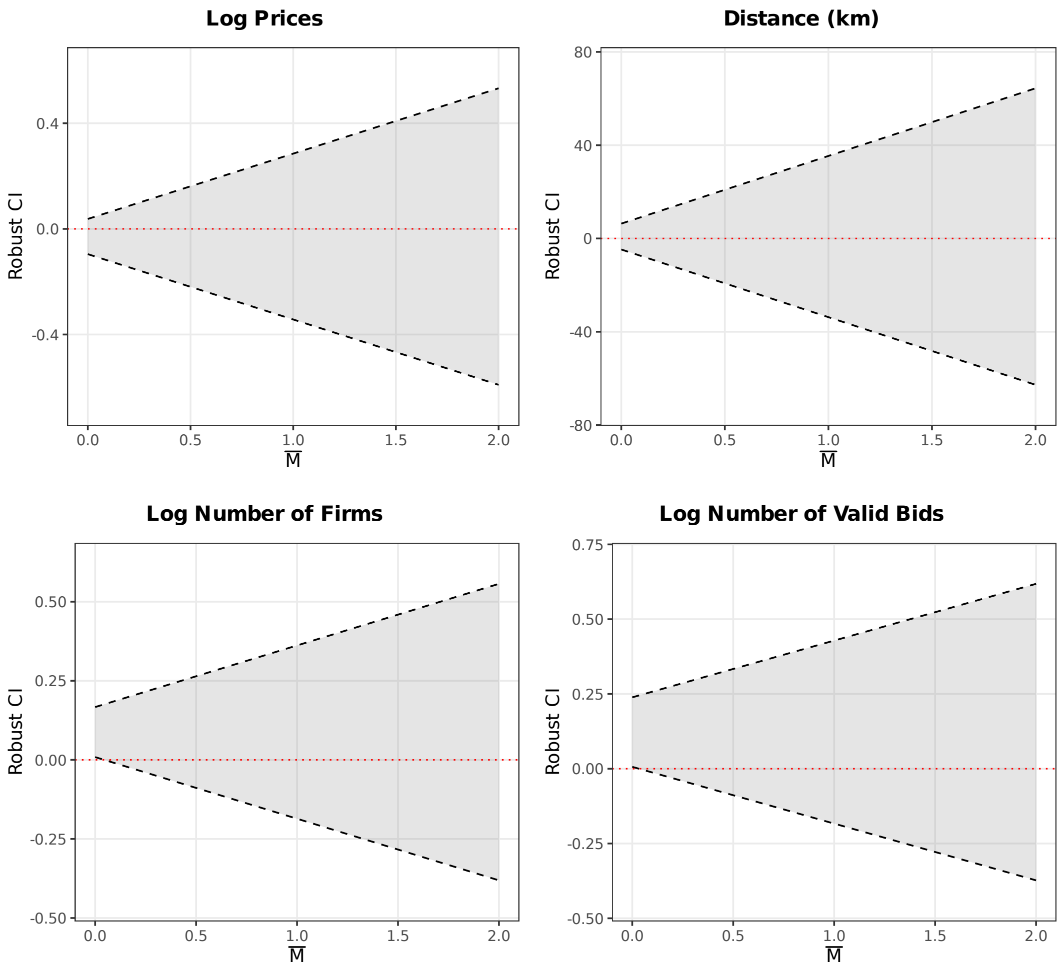

Figure A.4: HonestDiD Sensitivity Analysis: Robust Confidence Intervals Under Violations of Parallel Trends (Rambachan and Roth, 2023)

Table B.6: Lee (2009) Bounds: Sample Selection Correction

	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.0460***	-0.1338***	27.0609***	-19.0176***
	(0.0131)	(0.0114)	(2.3804)	(2.3672)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1886	0.1951	0.0015	0.0017
Trimming proportion		0.0882		
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** p<0.01, ** p<0.05, * p<0.1.

Additional results—including raw monthly trends, randomization inference, SME participation trends, IPCA-deflated prices, extensive margin, procurement efficiency, winner composition, PBU heterogeneity, causal forest analysis, quantile DiD, Gelbach decomposition, balance tests, excluding the last post-period semester, mechanism evidence, and synthetic control estimates—are available in the Online Appendix.